AMERICAN AND ENGLISH LOCOMOTIVES.

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SPECIFICATION OF ENGLISH EXPRESS PASSENGER LOCOMOTIVE.

LONDON & SOUTHWESTERN RAILWAY.

TENDER FOR FOUR-WHEELS COUPLED BOGIE PASSENGER ENGINES AND TENDERS.

DRIVING AND TRAILING-WHEEL CENTERS.

THE wheel centers to be of good sound cast steel, of approved make, free from honeycomb and other defects. One wheel center out of 40 is to be tested to destruction under the following conditions :

The wheel center is to be raised in a running position and allowed to fall upon a solid foundation from the following heights : 10 ft., 15 ft., 20 ft., 25 ft., 30 ft.

Should any wheel center break at the two lower heights—viz., 10 ft. or 15 ft., and show defects on hard material, the Railway Company's Locomotive Superintendent or his Inspector shall have the power to reject the whole. The wheels to be inspected on the premises of the maker.

Tensile test pieces are to be taken from the wheel center to give a breaking strain of not less than 28 tons per square inch, with an elongation of not less than 20 per cent. in 2 in.

Each wheel center is also to be tested by being allowed to fall in a running position a distance of 4 ft. 6 in. on to a wooden block without showing any signs of defect.

All the wheel centers must be bored and turned, and have keyways cut strictly to template, so that they shall be exactly alike, and each wheel must be forced on the axle before the tire is shrunk on by a hydraulic pressure of not less than 80 tons. The rims must be correctly turned to gauge to receive the tires, and the whole wheel trimmed up so that the surfaces and lines are all fair and true. The wheel centers are to be turned to a diameter of 6 ft. 7 in., the rims are to be 4¾ in. broad, 2¾ in. thick at center, to have 22 spokes 2½ in. thick at the boss, and 4 in. deep, and at the rims 1⅝ in. thick × 3½ in. deep. The bosses are to be bored out parallel to a diameter of 9½ in., and are to be 1 ft. 5 in. diameter. The cranks for the coupling-rods are to be cast solid with the bosses, 13-in. centers, and bored out parallel to a diameter of 5¼ in. to fit the coupling-rod crank-pins. The crank-pin holes are to be bored in a suitable quartering machine. The balance weights to be cast solid, and to be different for the driving and trailing wheels. Care to be taken that each wheel is cast with its proper balance weight. Generally the wheel centers must be as shown on the drawing.

BOGIE-WHEEL CENTERS.

The bogie-wheel centers are to be of good sound cast steel of approved make ; quality, manufacture and tests same as specified for driving and trailing-wheel centers. Each wheel center to be turned to a diameter of 3 ft. 3½ in., the rims are to be 4⅝ in. broad, 2½ in. thick at center, to have 10 spokes 1⅞ in. thick at the boss and 4 in. deep, and at the rims 1⅝ in. thick and 3¼ in. deep. The bosses are to bored out parallel to a diameter of 7 in., and are to be 1 ft. in diameter. The wheel centers must be bored and turned strictly to template, so that they shall be exactly alike. Each wheel center must be forced on the axle by a hydraulic pressure of not less than 70 tons. The wheel centers are to be fixed to the axles without keys.

TIRES.

The tires are to be of the best cast steel, manufactured by Vickers & Co., and are to be tested at the works of the maker in the following way :

Each tire is to be guaranteed to stand without fracture the repeated falling of a 1-ton monkey from a clear height, first blow 10 ft. ; second, 15 ft. ; third, 20 ft. ; fourth, 25 ft. ; fifth, 30 ft., and so on. Any tire which cracks or breaks before it has deflected one-sixth of its external diameter is to be rejected.

Test pieces are to be machined cold out of the tire without reheating the steel or treating in any way beyond cold machining for tensile test. The minimum tensile strength to be 44 tons per square inch, and to have an extension of not less than 15 per cent. in 2 in. A suitable and sufficiently large piece is to be sent to nine Elms for testing in a similar manner.

The contractor shall require the maker to provide at his own expense one additional tire for each 50 ordered, to be selected from the bulk by this Company's Locomotive Superintendent or his Inspector, and to be tested in his presence by the maker in the manner before described.

In the event of one tire cracking or breaking, or failing to stand the test, the company to have the power to reject the whole.

The number of the charge is to be stamped on each tire, and in the event of there being more than one charge in every 50 tires, a tire shall be selected from each charge and tested.

The maker's name and date of manufacture is to be stamped on each tire.

All the tires are to be 3 in. thick, of the form shown on drawing, and to be secured to the wheels with a lip and steel set screws 1⅛ in. in diameter, 11 threads per inch. Each tire to be bored to gauge before being shrunk on the wheel center. Each tire to be accurately turned, so that the diameters and thickness shall be exactly similar.

AXLES.

All the axles must be of the very best cast steel, manufactured by Vickers & Co., and must be stamped with the maker's name and date of manufacture. Test pieces are to be made, giving a tensile strength of not less than 28 tons and not more than 32 tons per square inch, with an elongation of not less than 25 per cent, in 2 in. ; a piece of a suitable length, 1¼ in. square, is to be bent double when cold without showing any signs of failure. The bogie axles to have centers of bearings 3 ft. 7 in., 5⅛ in. diameter, 10 in. long. The driving and trailing-axles to have centers of bearings 3 ft. 9¾ in., 8 in. diameter, 9 in. long. All axles are to be as shown on drawings.

DRIVING AND TRAILING AXLE-BOXES.

The driving and trailing axle-boxes to be as shown on drawing, of the best gun-metal, and to have bearing surfaces of Dewrance's anti-friction metal : keeps to be of cast iron. The axle-boxes to have lubricating pads as shown. There is to be only one groove in the crown of the axle-boxes, with the lubricating holes leading into it. The axle-box bearings to be $\frac{1}{16}$ in. shorter than the axle journal, to give clearance. The axle-boxes must have $\frac{3}{32}$ in. side play on each of the guides. Each axle-box must be made to gauges and must be duplicates of each other.

BOGIE AXLE-BOX.

The bogie axle-box to be of the best gun metal ; keeps to be of cast iron, to have bearing surfaces and provision for lubrication, as shown on drawing. The axle-box bearings to be $\frac{1}{16}$ in. shorter than the axle journal to give clearance.

DRIVING AND TRAILING SPRINGS.

The springs are to be made of the very best quality of spring steel, manufactured from Swedish bar iron. Five per cent. of the bars to be tested at the works of the makers by the Railway Company's Locomotive Superintendent or his Inspector in the following manner : A piece to be cut from each bar 2 ft. 6 in. long, heated and bent round to a radius equal to 80 times the thickness of the bar, then hardened and tempered. The camber to be taken after it has been pushed straight once in the testing machine, after which the bar must be pushed straight six times without showing any further permanent set. The tensile strength of the bars to be not less than 45 tons per square inch, with an elongation of not less than 15 per cent. in 2 in. Manufacture and brand to be approved by the Railway Company's Locomotive Superintendent. The plates are to be truly fitted, tempered and stamped with the maker's name and date of manufacture. The plates to be prevented from shifting side or end ways by nibs stamped upon them. Care must be taken that the nibs formed on the plates fit the slots properly. The buckles are to be sound forgings and are to fit the springs accurately, and are to be well secured to them, the buckles to be prevented from shifting on the springs by short wrought-iron pins, driven while hot through holes in the top and bottom of the buckle, and into a hole in the top plate and a recess in the bottom plate, as shown on drawing. The springs are to consist of 12 plates ½ in. thick and 5 in. broad, to a span of 4 ft., and to have adjustable hangers at the end and solid hangers in the center. Each spring must be thoroughly tested before being put in its place, by being weighted with 11 tons, and on the removal of this weight the spring must resume its original form.

BOGIE SPRINGS.

The material, workmanship, method of construction and testing of the bogie springs must be the same as for the driving and trailing springs. The bogie springs are to consist of 14 plates ½ in. thick, 5 in. broad, to a span of 3 ft. 11⅛ in.